WORKSHEET 7.2

Checking a Proposed Solution

Unit 7: Differential Equations

Key ideas

- A **solution** of a differential equation is a function that makes the equation true for every x in an interval.
- To check: compute the derivatives, substitute into each side *separately*, and compare.
- A **general** solution has an arbitrary constant; an initial condition selects a **particular** solution.
- A constant function $y = c$ is a solution exactly when it makes the right-hand side zero.

Worked examples

Example 1 · A direct check

Show that $y = 3e^{2x} + 1$ is a solution of $\dfrac{dy}{dx} = 2y - 2$.

Solution. Left side: $6e^{2x}$. Right side: $2\left(3e^{2x} + 1\right) - 2 = 6e^{2x}$.

The sides are equal for all x.

Example 2 · Finding an exponent

For what values of k is $y = e^{kx}$ a solution of $y'' - y' - 6y = 0$?

Solution. $e^{kx}\left(k^2 - k - 6\right) = 0$, and $e^{kx} \neq 0$.

$k = 3$ **or** $k = -2$**.**

Example 3 · Constant solutions

Find the constant solutions of $\dfrac{dy}{dt} = y^2 - 4y + 3$.

Solution. $(y - 1)(y - 3) = 0$.

$y = 1$ **and** $y = 3$**.**

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Practice

CORE

1. Is $y = e^{-x}$ a solution of $y' + y = 0$?

2. Is $y = x^2$ a solution of $y' = 2y$?

3. Find the constant solution of $\dfrac{dy}{dt} = 5 - y$.

4. The general solution of a differential equation is $y = Ce^x$. Find the particular solution through $(0, 7)$.

EXAM LEVEL

5. Show that $y = \sin(3x)$ is a solution of $y'' + 9y = 0$.

6. For what values of m is $y = x^m$ a solution of $x^2y'' - 12y = 0$ for $x > 0$?

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7. Show that $y = -\dfrac{3}{x^3 + C}$ is a solution of $\dfrac{dy}{dx} = x^2y^2$ for every constant C, and find C if the solution passes through $(0, 1)$.

8. Every function $y = A\cos(2x)$ satisfies $y'' + 4y = 0$. Find A if $y(0) = 5$.

Question 9 · Free-response practice

Let $y = xe^{-x}$.

(a) Find $\dfrac{dy}{dx}$.

(b) Show that y is a solution of $\dfrac{dy}{dx} + y = e^{-x}$.

(c) Find $\dfrac{d^2y}{dx^2}$, and show that y is also a solution of $y'' + 2y' + y = 0$.

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Worked solutions

1. Yes: $y' = -e^{-x}$, so $y' + y = 0$.
2. No: $y' = 2x$ but $2y = 2x^2$.
3. $y = 5$.
4. $C = 7$: $y = 7e^x$.
5. $y'' = -9\sin(3x)$, so $y'' + 9y = 0$.
6. $m(m-1) - 12 = (m-4)(m+3) = 0$: $m = 4$ or $m = -3$.
7. $\dfrac{dy}{dx} = \dfrac{9x^2}{(x^3+C)^2} = x^2 \cdot \dfrac{9}{(x^3+C)^2} = x^2y^2$. Through $(0,1)$: $-\dfrac{3}{C} = 1$, so $C = -3$.
8. $y(0) = A = 5$.
9. (a) $e^{-x} - xe^{-x} = (1-x)e^{-x}$.
   (b) $(1-x)e^{-x} + xe^{-x} = e^{-x}$.
   (c) $y'' = -e^{-x} - (1-x)e^{-x} = (x-2)e^{-x}$. Then $y'' + 2y' + y = \big[(x-2) + 2(1-x) + x\big]e^{-x} = 0$.

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